Final Project
Design Document

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Introduction

Project Functionality

The play area will consist of a square board (16x16) consisting of 256 tiles. The snake will move continuously in one of four directions (Up, Down, Left, Right) until the player switches the direction using WASD or arrow keys. There will be a single food item starting on a random tile, it will appear on a different tile after it has been eaten by the snake. Only the head of the snake can eat the food item, and the tail will grow by 1 tile once it has been eaten. The game will end if the head of the snake collides with the border wall or itself (tail). The amount of food items consumed in a game will contribute to the player's score. Each food item will add 1 point and after the game ends a high score will be presented beside the total amount of food consumed (score). The game over screen will also show a message when the snake dies and below it will be a restart button.

This is not a promise, but I want this game to be able to run forever. If it is possible and I have the time to implement it, I want the snake to be able to starve. As time passes, the snake tail will shrink by 1 unit. This time will have to be carefully calibrated in order to make it fun. This way the game can run forever, and food has more than one purpose. I believe time elapsed should contribute to score or be separate.

Design Process

At this moment the design process is still being tweaked, and the game is under development. A barebones version will be made to test and refine the code. Once the code is optimized and the gameplay works, the snake and background will be redesigned.

Project Development

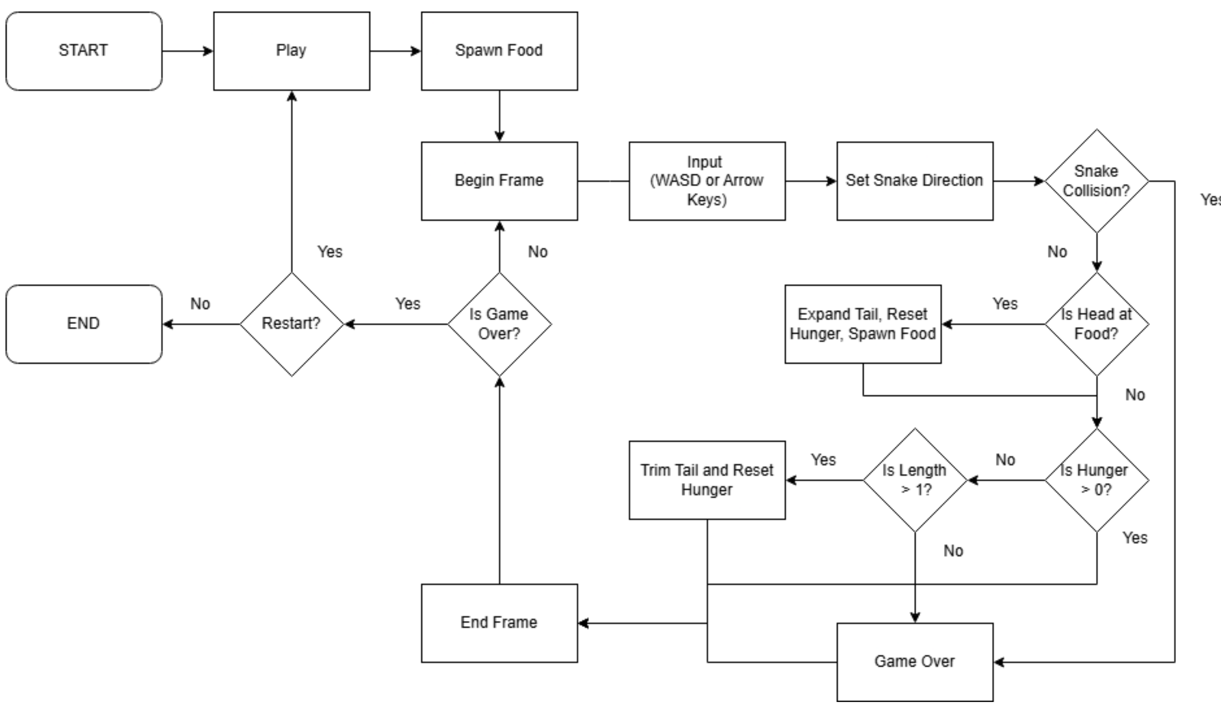
Pseudocode

START GAME TILE_SIZE = 64
BOARD_SIZE = 16

```
FRAME_RATE = 60
Construct game board (TILE_SIZE, BOARD_SIZE)
length = 3
direction = RIGHT
Spawn food ()
hunger_timer = 4
score = 0

LOOP WHILE game is running: IF user input (WASD or Arrow Keys):
Change direction
Calculate head = snake_head + direction

IF head is outside board OR inside snake body:
   END GAME

IF head is at food position:
   Increase score by 1
   Reset hunger_timer
   Spawn food ()
ELSE:
   Subtract 1 from hunger_timer

IF hunger_timer == 0:
   IF snake length > 1:
      Remove tail segment
      Reset hunger_timer
   ELSE:
      END GAME

Wait for next frame

END LOOP
```

This is a draft of the main code for the project, specifically the core game loop or algorithm. When the game starts the board is constructed and variables are assigned. The plan is that the code will loop every frame using functions that have already been defined. I want to use classes or some sort of object-oriented concept.

Flowchart



This is the flowchart for the game without the score mechanic for simplicity.

UML Diagram

This is for your UML diagram**.** Please provide your UML diagram (if you need to create one).

Do we need to create one?

Requirements

1. Board Size / Play Area: The game should be played in a window containing at least a 10-by-10 tile set.
   Project does not currently meet this requirement; under development.

2. Snake Movement: The snake should move continuously in one of four directions (up, down, left, right). The player should control the direction using the arrow keys.
   Project does not currently meet this requirement; under development.

3. Snake Growth: Every time the snake eats a food item, it should grow longer by one unit.
   Project does not currently meet this requirement; under development.

4. Food Generation: Food should appear randomly on the game screen after being eaten by the snake.
   Project does not currently meet this requirement; under development.

5. Collision Detection: The game should end if the snake collides with the walls of the screen or its own body.
   Project does not currently meet this requirement; under development.

6. Game Over and Score: Display a Game Over message when the snake dies.
   Project does not currently meet this requirement; under development.